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Palaeomagnetism of the Ordovician Moulin de Chateaupanne formation, Vendee, western France

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Summary. A palaeomagnetic study of the Moulin de Chateaupanne formation reveals two characteristic components of magnetization. The first one, isolated by thermal treatment at intermediate temperatures, is strongly resistant to chemical leaching. Its direction ($D = 217°$, $I = +25°$, $\alpha_{95} = 5.5$) and the corresponding pole position (Lat = 21°S, Long = 320°E) are typical of the Hercynian overprints in the Armorican Massif. The second component, carried by fine-grained haematite, presents a steep downward direction ($D = 228°$, $I = +81°$, $\alpha_{95} = 6°$; pole position Lat = 34°N, Long = 343°E) already recognized in other Ordovician rock units from the same area. It is resistant to high-temperature demagnetization treatment and is believed to be of chemical origin, as assessed by a conglomerate test. A mutual consistency test with the nearby Thouars Massif suggests that the tilt of the formation has a sedimentary origin, and that the Thouars Massif has not suffered significant tilt during the Hercynian orogenic events.

Key words: later magnetism, Ordovician, Armorican Massif, redbeds

Introduction

The Armorican Massif (Western France) is very rich in Lower Palaeozoic redbeds. Usually they are unfossiliferous and their stratigraphic age is poorly known, leading sometimes to controversial interpretations (see, e.g. Perroud *et al.* 1985). A palaeomagnetic study of most of these redbeds has been carried out by Duff (1979) in a broad survey of the whole Armorican Massif. One of the sites studied by Duff (LV) is the Moulin de Chateaupanne. It is in fact an exception to all other units: it is a well stratified detrital formation, which is unfolded and unmetamorphosed, gently dipping to the NW and discordantly overlying Brioverian greywackes. It is composed of reddish to yellowish sandstones, breccias or siltstones, including microconglomeratic levels with Brioverian pebbles. It has been attributed an Arenig age on the basis of brachiopod and conodont faunas (Blaise, Cavet & Lardeux 1970; Cavet *et al.* 1979).

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Therefore, in our effort to obtain lower Palaeozoic palaeomagnetic poles for the Armorican regions, and as an alternative to the approach adopted by Duff of combining results from all redbeds studied, we have chosen this formation for a detailed palaeomagnetic study. Furthermore, sedimentary rocks offer the advantage of a well defined palaeohorizontal (bedding), although the mechanism of remanence acquisition is not very well understood. By contrast, a thermoremanent magnetization (TRM) is usually present in igneous rocks and is easily reproduced in the laboratory, but indications of tilt are generally lacking. A combination of these two kinds of data is highly desirable, and that is one of the aims of this study. In a related study we have recently obtained an Ordovician palaeomagnetic pole determination from the Thouars hypovolcanic massif (Perroud & Van der Voo 1985) situated a few tens of kilometres from Chateaupanne. These two studies provide the opportunity for mutual consistency tests and the reliability of both results will be enhanced by the comparison.

Sampling and methods

The 'serie du Moulin de Chateaupanne' is known only in a few localities on the southern margin of the Ancenis syncline (Fig. 1); it lies as patches on the Brioverian basement of the 'Série des Mauges', isolated from the folded Palaeozoic sequence of the Ancenis syncline by Hercynian faults. Furthermore, the red coloration of the formation is not systematic (Blaise *et al.* 1970). Only two sites have been found suitable for palaeomagnetic investigation (labelled MC and DC in Fig. 1): first, the Chateaupanne quarry (MC), where a complete cross-section of the formation is available. This has been widely sampled with 25 cores drilled in the field plus nine oriented hand-samples. Moreover, a large hand-sample of breccia has been collected for a conglomerate test. At the other site, les Deux-Croix (DC), three oriented hand-samples have been collected.

Remanent magnetization measurements were made with the Schonstedt DSM-1 computer-assisted magnetometer. Standard cleaning procedures were performed (using Schonstedt equipment) on all specimens, including af, thermal and chemical progressive demagnetizations. In addition, isothermal remanent magnetization (IRM) acquisition experi-

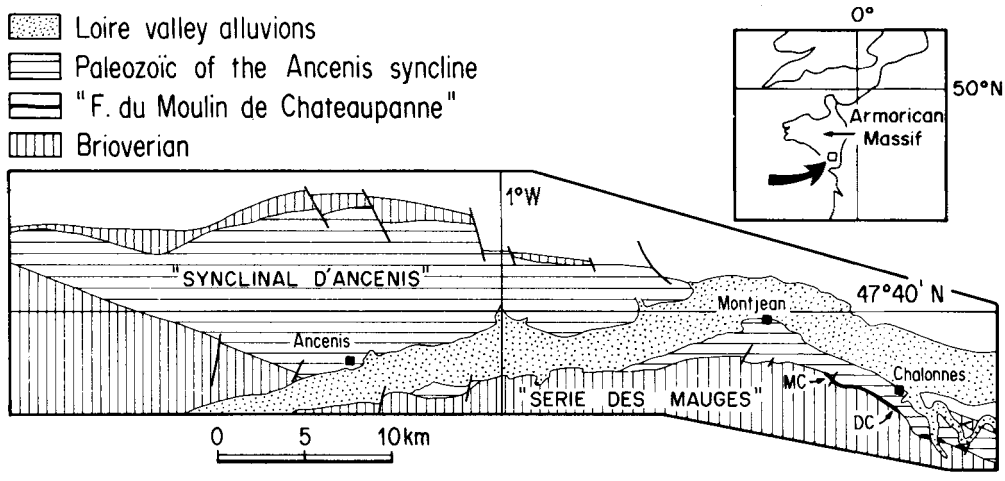


Figure 1. Schematic geological map of the Ancenis syncline, with the localities of our sampling sites (MC: Moulin de Chateaupanne; DC: les Deux-Croix). After Blaise *et al.* (1970).

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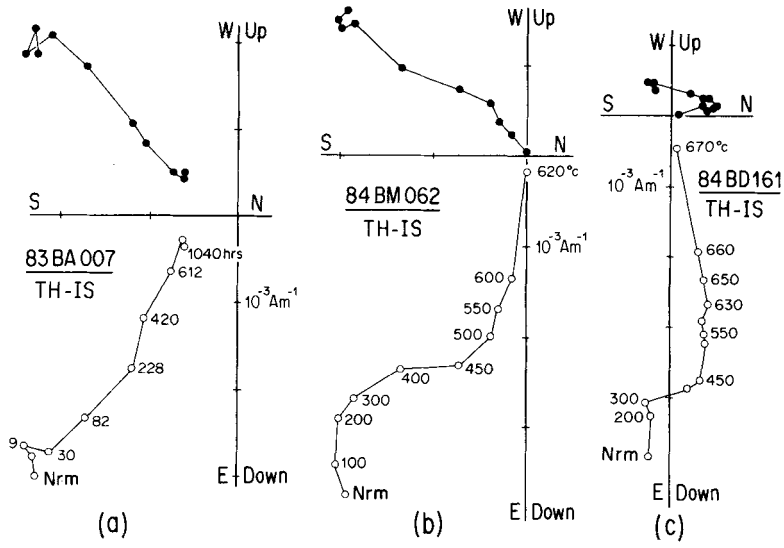


Figure 2. Zijderveld diagrams of demagnetization results. Full (open) dots are for the projection on to the horizontal (vertical) plane: (a) chemical demagnetization; (b) thermal demagnetization of a sample from the Chateaupanne quarry; (c) thermal demagnetization of a sample from les Deux-Croix.

ments were performed on several pairs of leached and unleached specimens, and thermomagnetic curves were produced, using a Brucker Curie balance. Demagnetization results were displayed on Zijderveld (1967) diagrams and graphically interpreted.

Results of the palaeomagnetic analysis

NRM intensities are fairly uniform throughout the whole specimen set, in the range 10E-3 to 10E-2 A m^{-1}. Directions are reasonably well-grouped (in the SW, moderately inclined downwards). Only a few specimens from 'Les Deux Croix' show directions which seem to have deviated towards the present-day field direction.

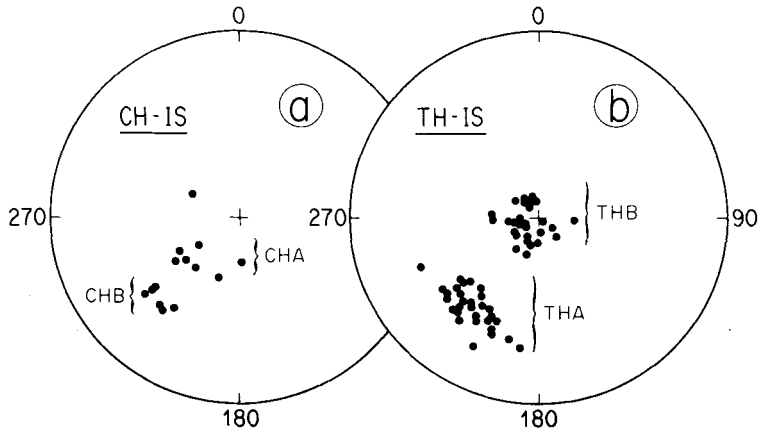


Figure 3. Lower hemisphere stereographic projections of the characteristic components of magnetization (not corrected for tilt): (a) results from chemical demagnetizations; (b) results from thermal demagnetizations. Symbols are explained in the text.

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Af demagnetization up to 100 mT has little effect on the remanent magnetization, as is generally the case in redbeds where haematite is the main magnetic carrier. Upon chemical leaching in HCl acid (11 specimens), the intensity of magnetization decreased rapidly during the first 200 hr (normality 6 N), then goes down to a minimum value characterized by a flat portion of the intensity decay curve for leaching times above 600 hr (normality increased to 12 N). About 20 per cent of the initial magnetization remains unaffected after leaching for 1000 hr. The coloration of the samples changes progressively from red to grey during the treatment. Characteristic directional changes during the procedure (Fig. 2a) usually correspond to a lowering of the inclination while the declination remains stable. The Zijderveld diagrams show a linear trajectory bypassing the origin, until a 'stable end point' is reached when leaching ceases being efficient. This behaviour indicates a multivectorial origin of the magnetization. Two groups of directions can be obtained from these experiments. One (CHA), including all the components removed by leaching, is estimated from the initial linear segment of the demagnetization diagrams; the *in situ* mean direction is $216^\circ/+54$, $k = 32$, $N = 7$. The other (CHB) is defined by the end-points and has an *in situ* mean direction $221^\circ/+27$. $k = 152$, $N = 6$. The individual data are plotted on stereographic projections in Fig. 3(a).

Thermal demagnetization (36 specimens) confirms the multivectorial nature of the NRM: up to 300°C, a small vertical component is removed, very likely of viscous origin (Fig. 2b, c): then a southerly shallow component is destroyed below 450–500°C; finally the magnetization decays towards the origin of the Zijderveld diagrams, revealing a steep component of magnetization, which is stable in the range 600–670°C. This behaviour is observed as well at the Chateaupanne quarry (Fig. 2b) as at les Deux-Croix (Fig. 2c). It should be noted that one hand-sample from the last site revealed a present-day field component at temperatures below 500°C and an anomalous easterly direction at higher temperature; it has been excluded from further consideration. The *in situ* mean directions for the last two components are: 215/+25, $k = 42$, $N = 32$ for the intermediate (THA), and 234/+82, $k = 35$, $N = 29$ for the high unblocking temperature (THB) component. Corresponding data are shown in Fig. 3(b). These results are in good agreement with those mentioned by Duff (1979, table 3, p. 353), for site LV, obtained from two specimens.

Analysis of magnetic carriers

The maximum unblocking temperature identified during thermal demagnetization (670°C), together with the resistance of the remanence to af demagnetization, implies that haematite carries a significant part of the magnetization, including the high-temperature component. However, we cannot exclude contributions from other magnetic minerals. We have therefore conducted experiments on pairs of leached and unleached specimens to attempt to identify the carriers of the two components (removed and residual magnetizations) isolated by the chemical treatment.

Weakly magnetized sediments need a very careful processing of the thermomagnetic cycles. In particular, as we are using a vertical balance, it is necessary to control weight loss. Heating was therefore conducted in the following manner: the temperature was increased in 50°C steps; when stabilization was obtained, the applied magnetic field (about 0.8 T) was turned off, and the balance zeroed; then the field was turned on and the magnetic strength measured. The heating curve was thus obtained in a discrete fashion. Upon cooling, measurement was continuous. We notice on Fig. 4(a) the excellent overlap between the discrete heating curve (black dots) and the continuous cooling curve, showing the reversibility of the processes (experiments in air). From the same specimen, identical weights of leached and

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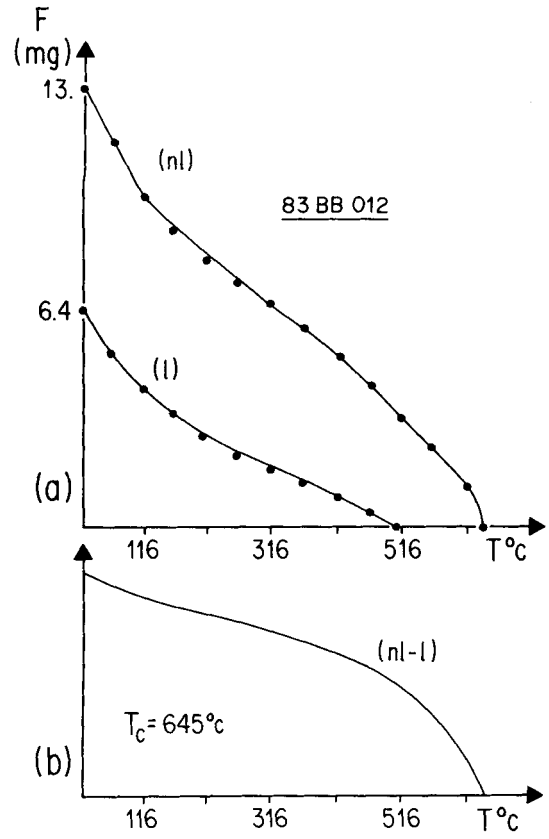


Figure 4. (a) Thermomagnetic curves obtained from leached (l) and not leached (nl) fragments of the same specimen. The dots represent the discrete measuring points during heating. The continuous curve corresponds to cooling. (b) Curve obtained by subtraction (nl – l) between the two experimental ones, characterizing the magnetic constituent dissolved during leaching.

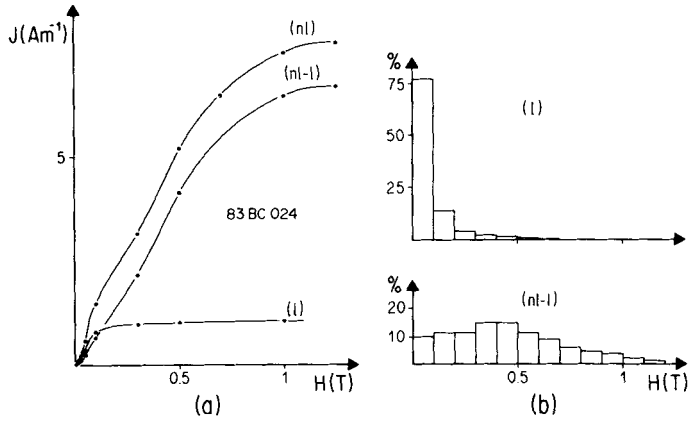


Figure 5. (a) IRM acquisition curves for two pieces of the same sample, one having been leached (l) and the other not (nl). The curve labelled (nl – l) was obtained by subtraction. The dots correspond to the measuring points. (b) Coercivity spectra representing the dissolved (nl – l) magnetic fraction and the residual (l) one.

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fresh rock powders were analysed (Fig. 4a). The difference of these two curves has been constructed and is shown in Fig. 4(b); the result is therefore representative of the magnetic minerals dissolved by the acid. It is typical of a ferromagnetic constituent, with a Curie point around 650°C, that is nearly pure haematite. On the contrary, the leached specimen does not reveal any trace of this constituent (Fig. 4a). The concavity below 200°C can be attributed to paramagnetic minerals, or to a low Curie temperature mineral, such as ilmenite; above 200°C, a second Curie temperature appears (about 520°C), which characterizes the magnetic carrier unaffected by chemical leaching.

A similar analysis has been performed with progressive IRM acquisition experiments. A pair of leached and unleached pieces of the same sample were given IRMs up to 1.25 T, and the behaviour of the dissolved magnetic components was determined taking the difference between the two IRM acquisition curves (Fig. 5a). Coercivity spectra were calculated for both dissolved and residual magnetic grains (Fig. 5b): the first is typical of fine-grained haematite (Dunlop 1972), while the second shows a very large acquisition in fields lower than 0.1 T and a weaker contribution up to 0.5 T. The interpretation of this last magnetic phase is more ambiguous: large grain haematite (specularite) or even magnetite are plausible candidates.

Interpretation of the magnetization components

The intermediate unblocking temperature (THA) and the chemically undissolved (CHB) components show identical directions. Moreover, the Curie temperature of 520°C obtained from the J_s-T curve of the leached sample is very consistent with the temperature range of stability for that component. On the contrary, the high temperature steep direction (THB) does not fit with either component obtained by chemical demagnetization (CHA or CHB). However, since CHA has a direction intermediate between THA and THB (Fig. 3) we argue that chemical demagnetization has not completely isolated the characteristic magnetization. The apparent components removed by leaching (CHA) are a combination of two (or more)

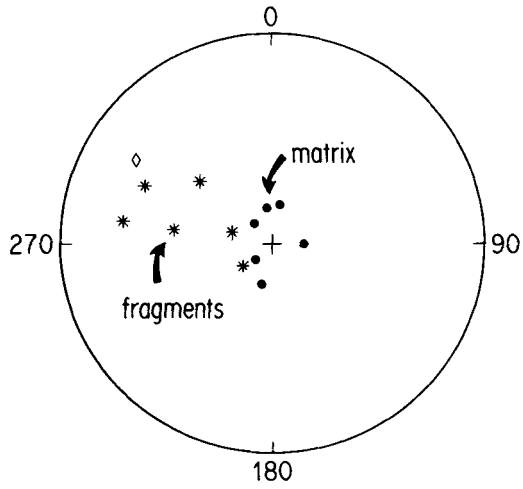


Figure 6. Results of the conglomerate test: the dots are the NRM directions of the matrix samples, while the stars (lower hemisphere) and the diamond (upper hemisphere) were obtained from samples drilled in the fragments of Brioverian greywackes in a large hand-sample. Stereographic projection, in a sample reference frame.

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magnetizations. They will not be considered further here and we will use rather the thermal results. Another conclusion is that the fine-grained haematite (high coercivity and Curie temperature around 650°C) carries both the high-temperature steep component and a part of the shallow component, while the undissolved magnetic grains carry only the shallow component.

Further constraints can be obtained on the age of acquisition of these components from the geological background. First a conglomerate test has been performed on a large hand-sample containing pebbles of Brioverian meta-sediments. The size of the pebbles (a few centimetres) was too small to allow standard sampling. We have therefore drilled 8 mm cores in the matrix and in a set of pebbles. The magnetic moment was very low and a cryogenic magnetometer was used to make the measurements. Because of the weak intensity of magnetization, no demagnetization was possible and only NRM results are reported here. The results are displayed in Fig. 6 and it appears that the pebble magnetization is not random. On the contrary it presents a trend towards the matrix magnetization direction. We therefore propose that the acquisition of magnetization in the matrix is younger than the sedimentation of the pebbles and we assign a post-depositional (chemical) origin to the matrix magnetization, carried in large part by the fine-grained haematite pigment.

Another useful way to date the magnetization acquisition in sediments is the application of the Graham (1949) fold test. Unfortunately, we have not been able to find sites with divergent bedding planes, and this prevents us from obtaining a conclusive test. However, bedding attitude varies slightly from place to place in the Moulin de Chateaupanne quarry, and between the quarry and les Deux-Croix To compare the dispersion of palaeomagnetic data before and after tilt correction, we have divided the area of the quarry into nine sub-sites. Within each sub-site, the bedding is uniform, and mean directions have been computed for both components (Table 1). These directions were then used as the entries in a statistical test for comparing the dispersion parameter k. For the shallow component, both chemical

Table 1. Results of the palaeomagnetic analysis.

SITE	Bedding		Component of magnetization 1				Component of magnetization 2			
	Strike	Dip	N	D/I (IS)	D/I (TC)	k	N	D/I (IS)	D/I (TC)	k
Moulin de Chateaupanne quarry :										
Group 1	192	14	6*	216/24	221/18	244	3	238/79	262/67	32
Group 2	183	9	1	231/24	233/17	-	1	261/73	265/64	-
Group 3	208	13	3*	209/23	214/22	33	2	140/78	231/85	14
Group 4	196	18	7*	223/32	231/23	159	4	258/83	278/65	40
Group 5	251	6	3	229/21	231/23	32	4	177/79	194/85	44
Group 6	190	7	2	210/20	212/18	36	1	205/75	226/72	-
Group 7	182	9	5	212/27	216/22	49	4	258/76	263/67	28
Group 8	176	9	4	213/28	216/22	55	4	231/78	246/70	103
Group 9	195	15	3	222/23	227/16	294	2	210/70	240/62	62
Les Deux Croix :										
	221	19	4	202/22	210/27	49	4	332/80	318/61	1014
Mean IS :			10	217/25		77	10	228/81		65
Mean TC :			10		221/21	84	10		261/72	38

Notes : The Moulin de Chateaupanne quarry samples were divided in 9 groups for which bedding is uniform. A star (*) in the N (number of specimens) row indicates that both thermal and chemical results were used. Otherwise, only thermal results were used. Component of magnetization 1 refers to directionnal groups CHB and THA ; component 2 to THB (see text).

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and thermal results were used (CHB + THA); we obtained k values of 77 and 84 respectively before and after the tilt correction ($N = 10$). These values are essentially identical which is not surprising since the tilt is small and the direction of magnetization close to the strike direction (SW). The direction of that component does not vary by more than a few degrees after the tilt correction. On the contrary, the dispersion parameter of the high-temperature direction (THB) is larger before the correction ($k = 65$ before, 38 after, $N = 10$), but the k ratio is not high enough for the result of this test to be statistically significant.

Comparison with the Thouars Massif palaeomagnetic results

We have recently studied (Perroud & Van der Voo 1985) gabbroic samples from the hypovolcanic Thouars Massif situated in the immediate neighbourhood of Chateaupanne. The Massif has been dated at 444 ± 9 Ma by a Rb/Sr isochron (Bernard-Griffiths & Le Metour 1979): our study revealed a nearly vertical primary magnetization, with both normal and reverse directions. A secondary magnetization, found on the border of the massif where Hercynian deformation has affected it, has been isolated with a SW and shallow direction. We have proposed an orogenic origin for this overprint. The most critical point in this study was the estimation of a possible tilt of the whole massif subsequent to the magnetization acquisition. Due to the spatial and temporal proximities of both rock units, we think that we now have the opportunity for a mutual consistency test.

There are strong similarities in the results of both studies, although the rock type is very different. Two components of magnetization were found in both cases: one is vertical, corresponding to high unblocking temperatures; the other is shallow downwards and south-westerly, corresponding to intermediate unblocking temperature (below 500°C). The directions of both soft components are identical (Fig. 7) and their temperature ranges of stability similar. We therefore conclude that they have the same origin. As already concluded from the Thouars gabbro, we propose that they are overprints acquired during the Hercynian orogeny. The similarity of the unblocking temperature suggests a thermal nature. Corrections for tilt do not affect these conclusions.

The directions of the high-temperature components are also quite close to each other

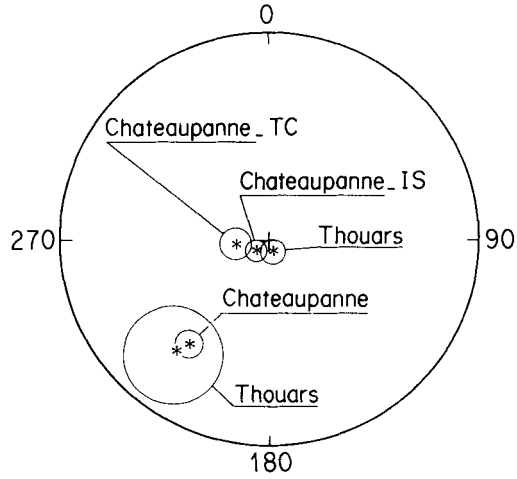


Figure 7. Comparison of the mean palaeomagnetic directions obtained in this study with those obtained from the Thouars Massif results. Stereographic projection, lower hemisphere.

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(Fig. 7). In the Thouars gabbro the magnetization is carried by fine-grained magnetite, with unblocking temperatures just below the Curie point. This has been interpreted as the primary magnetization of the gabbro. In the Moulin de Chateaupanne redbeds, the corresponding component is carried by fine-grained haematite, also with unblocking temperatures in the proximity of the Curie point. Furthermore, a similar steep high-temperature component has been found in redbeds considered to be a lateral equivalent, in the Buçaco syncline (Perroud 1982); we believe that this steep component is the initial magnetization of the studied formations, although its age is difficult to assess since it has a chemical origin. Due to the resemblance with the gabbro primary direction, we are inclined to propose an Ordovician diagenetic acquisition.

This mutual consistency test can also be developed to check the influence of tilt on both results. In Fig. 7, both uncorrected and tilt corrected results are displayed, and compared to the Thouars result. It appears that the correction pulls the remanence direction for the redbeds away from that of the gabbro. While the two vectors cannot be distinguished before, they became significantly different after application of the correction. Therefore, our mutual consistency test suggests (1) that the Thouars gabbro has not suffered significant tilt and (2) that the redbeds magnetization was acquired after the tilt.

Conclusions

Our palaeomagnetic study of the Moulin de Chateaupanne formation indicates the presence of two characteristic components of magnetization:

(1) The first has a south-westerly shallow downwards direction and is isolated by thermal treatment between 300 and 500°C. It also corresponds to the residual magnetization after 1000 hr of chemical leaching. It is carried in part by a magnetic fraction with a Curie point about 520°C and coercivity lower than 0.5 T, for which we assume a detrital origin. However chemical leaching significantly destroys that component, and it is quite possible that the fine-grained haematitic pigment carries another part. Tilt correction does not significantly affect that component (neither dispersion nor direction). Its direction is identical with that of the Hercynian overprint of the Thouars gabbro and also similar to all Carboniferous results from the Armorican Massif (Perroud *et al.* 1985). We relate its acquisition to the thermal events of the Hercynian orogenesis in the southern part of the Armorican Massif. We regard the *in situ* mean direction (217/+25, $\alpha_{95} = 5.5$), as the better estimate of that component. The corresponding palaeomagnetic pole position is Lat = 21°S, Long = 320°E.

(2) The other component has a steep downwards direction, and is isolated by high-temperature treatment (above 500°C, up to 670°C). It is carried exclusively by the fine-grained haematitic pigment, with a high coercivity and a high Curie temperature. It is believed to have a chemical post-depositional origin, as indicated by the results of a conglomerate test. The resemblance with the Thouars gabbro primary magnetization suggests a diagenetic acquisition, in the Ordovician. However, this component appears to be younger than the tilting since the dispersion tends to increase, and the mean direction diverges from that of the Thouars Massif, after correction for tilt. These apparently contradictory conclusions may be justified from a consideration of the origin of tilt in the Moulin de Chateaupanne formation. According to Blaise *et al.* (1970), the Brioverian basement of the formation has been affected by faulting, accompanied by westwards rocking motions in Hercynian times. On the contrary, Diot (1980) proposed that sedimentation took place on a slope, whose angle decreased during the deposition, corresponding with the filling of a subsiding basin. The anomalous north-westerly tilt of the formation is more easily explained

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by this last model. Obviously, the second hypothesis fits better our palaeomagnetic conclusions; it reconciles the Ordovician age with the post-tilting origin of the magnetization. We consider therefore that the *in situ* mean direction (228/+81, $\alpha_{95} = 6$) is the better estimate of that component. It corresponds to the following pole position: Lat = 34°N, Long = 343°E.

Finally, we should mention that the results of this study provide confirmation of the Thouars gabbro results, especially with respect to the tilt problem. The steep Ordovician directions have been recognized now in several places. A synthesis using all available Ordovician directions has been published recently, where implications of the palaeomagnetic results obtained from Western European Palaeozoic rocks are discussed (Perroud, Van der Voo & Bonhommet 1984).

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